5	**(a)**	**(i)**	$B = \mu_o nI$ $= \left(4\pi \times 10^{-7}\right)\left(\dfrac{400}{0.35}\right)(2.3)$ [B1] $= 3.3 \times 10^{-3}$ T	
	(b)	**(i)**	The component of velocity perpendicular to the magnetic flux density results in a uniform circular motion. The component of velocity parallel to the magnetic flux density / along the axis is constant. These two motions give rise to helical motion.	B1 B1
		(ii)	Magnetic force provides for centripetal force [B1] $Bqv_{\perp} = m\dfrac{{v_{\perp}}^2}{R}$ $R = \dfrac{mv_{\perp}}{Bq}$ $v_{\perp} = R\omega$ and $\omega = \dfrac{2\pi}{T}$ [C1] $T = \dfrac{2\pi m v_{\perp}}{v_{\perp} Bq}$ $= \dfrac{2\pi m}{Bq}$ $= \dfrac{2\pi\left(1.67 \times 10^{-27}\right)}{\left(3.3 \times 10^{-3}\right)\left(1.6 \times 10^{-9}\right)}$ [M1: subs] $= 1.9873 \times 10^{-5}$ $\approx 20 \times 10^{-6}$ s [A0] for reference: $v_{//} = 2287.4$ m s^{-1} $v_{\perp} = 3522.4$ m s^{-1} $F_B = 1.8598 \times 10^{-18}$ N $R = 0.011140$ m $\omega = 3.1616 \times 10^{5}$ rad s^{-1}	

		(iii)	time in solenoid, $t = \dfrac{\text{length of solenoid}}{v_{//}}$ $= \dfrac{0.35}{4200\cos 57}$ $= 1.5300 \times 10^{-4}$ s [C1]	
		(iv)	number of cycles, $N = \dfrac{\text{time in solenoid}}{\text{period of helical motion}}$ $= \dfrac{1.5300 \times 10^{-4}}{20 \times 10^{-6}}$ $= 7.6992$ (7 complete cycles) [A1]	
		(v)	Principle: $F_B = Bqv$ Since: Since the <u>magnetic flux density outside the solenoid is weaker</u> Therefore: <u>F_B or the centripetal force decreases</u>. Or since the radius is inversely proportional to magnetic flux density Resulting in an increase in radius.	M1 A1

6	(a)	Number of protons = 82	B1